

**Review Report of revised manuscript: BLC-2026-04-0069.R2 (IET Blockchain
Wiley) received 24/7/2026**

Paper title: Cryptographic Chain-of-Custody for Last-Mile Parcel Delivery: A Blockchain-IoT System Architecture with Discrete-Event Simulation Evaluation.

Final decision: **Accept** (with final touch comments between the authors & the auditors in the proofing phase)

Comments: (I leave handling them between the authors and the editors, no need for another review.)

S1-d)

-In phrasing the results in section 6.3, beware that the paper reader doesn't know about the previous draft versions; i.e., don't compare with or mention something regular readers can't see. You may replace it with "the best-case scenario" if our application TXs always get into a block once they arrive.

-You may also add another column for the proportion of events confirmed within 15s (for clarity: all what is explained in the text can be found in the table) and another row for the "best case scenario" for the sake of comparison to keep the text intact without a lot of changes.

-I agree with 3 ranges, could give a reader or a designer a hint if comparing different public permissionless blockchains not just Ethereum.

S1-g, W3)

-When you say: "*The abstract's previous claim of sub-15-second confirmation for 94% of events has been replaced with the corrected range*", this note on the results only exists in Table4 explanation not in the abstract and not in the conclusion.

-Section 6.2.1 cannot appear before section 6.2

TDR)

-The abbreviation should be defined before first use (Tamper Detection Ratio) in the text before the table instead of/in addition to the table title.

-TDR remained vague to me for a while; I think some reasons (to help protect other readers from the same unclarity) are:

- The wrong order of 6.2 and 6.2.1 which is before the other? If 6.2 is first then now the abbreviation TDR is defined before first use.
- Should not mention exact ratio when discussing the centralized case, it is not related exactly to it but to your generated attack mix in the simulation. When you move in to the blockchain case, call it 6.2.2, you either add a column for the expected centralized case values based on your simulation data or suffice with saying something like "for this simulation data TDR will be 19.9%."

S3-threat model)

-On passive attacks, should add a single statement that this is more problematic in permissioned blockchains

(as written block construction and choice in Ethereum as an example goes through a lot of randomization; the problem becomes questionable in permissioned ones because, like the one closed in 2022, IDs from addresses are

easier to know and 33% is easier to reach. The original problem as I meant it and I think the authors get it and mean it too in what they wrote: if a competitor saw your TX in the private mempool he/she can take an action on real life, like a better deal to the destination, before the TX is confirmed; this is what resembles MEV in concept but the gain here is a real life business deal rather than an arbitrage money).

You don't have to get deep into much details than is currently written, but at least add to your statement on Ethereum that "passive attacks become more significant in permissioned settings" and either compare to MEV or to the closed company in 2022.

-Again, I'll leave the follow up to the editors.

-Writing Style:

This is for the editors to think about and solve with the authors (maybe it is part of the highlighting to reviewers and not in the final draft)

- Some text parts of the threat model (A4,A5) are written all in Bold font; cannot remain like this when published.
- The symbolic last paragraph shows how all attacks can possibly affect a transaction; so, should not be mixed as part of A5, find a suitable subtitle for it or maybe an indentation for attack classes to show that this is a final comment on all of them.
- For the not so much edited text in A1-A3, although condensed in one paragraph, the writing style still seems like a slide in a presentation more than text in a manuscript. Here is an example for the editors to observe; the text in red is just a suggestion, you can negotiate with the authors what you prefer or leave it as it is if it's OK with you.

Adversary A1 (External Network Adversary): ^{The attacker is assumed to be able to perform} Capabilities: eavesdropping on communication channels between IoT devices and blockchain nodes, man-in-the-middle interception, message ^{however,} replay. Cannot compromise device secure elements or obtain private keys. This adversary represents attackers on public cellular networks or WiFi infrastructure. ^{Possible defenses include} Defenses: ECDSA signatures prevent message forgery; timestamp integration and sequence numbers prevent replay; AES-256-GCM provides confidentiality.

Adversary A2 (Malicious Insider): Capabilities: possesses valid application-layer credentials (API keys, OAuth tokens) as a legitimate supply chain participant (courier, warehouse operator, or system administrator). ^{Has physical access to parcels.} Objective: falsify custody records for financial gain (theft concealment, insurance fraud, counterfeit insertion). Cannot extract device private keys from hardware secure elements. Defenses: device-bound signatures prevent record falsification without physical device possession; on-chain immutability prevents retroactive record alteration. This is the primary adversary class against which blockchain verification provides its greatest advantage over centralized systems. ^{Seems better to follow has physical access to parcels, but cannot}

^{Here, we assume the attacker has extracted ...} **Adversary A3 (Compromised Device):** Capabilities: has extracted the ECDSA private key from one IoT seal device through physical attack (fault injection, focused ion beam, side-channel analysis). Can forge events from that specific device. Cannot compromise other devices. Defenses: per-device key isolation limits blast radius; DeviceRegistry revocation capability; fleet-wide statistical anomaly detection identifies compromised devices through behavioral profiling. This adversary class is explicitly acknowledged as a limitation of the cryptographic protocol; defense relies on hardware tamper resistance and operational monitoring.

Adversary A4 (Blockchain-Level, active): Capabilities: controls a minority of validator stake, enabling limited transaction reordering and selective censorship. The relevant threshold depends on the deployment: on Ethereum mainnet it is 33% of total staked ETH ^{As you can see the whole text is Bold here}